

JEE Main Physics Practice Test (PYQ Style)

Section A: 25 MCQs

1. A particle moves with constant speed in a circle. Its acceleration is
A) Zero B) Towards center C) Away from center D) Tangential
2. SI unit of electric flux is
A) N/C B) $\text{N}\cdot\text{m}^2/\text{C}$ C) C/m^2 D) V/m
3. Dimensional formula of Planck's constant is
A) ML^2T^{-1} B) ML^2T^{-2} C) MLT^{-1} D) ML^2T^{-1}
4. Escape velocity from Earth is proportional to
A) $\sqrt{R}$ B) R C) $1/R$ D) $1/\sqrt{R}$
5. The work done in an isochoric process is
A) Positive B) Negative C) Zero D) Infinite
6. A convex lens forms a real image. The image is always
A) Virtual B) Inverted C) Erect D) Same size
7. Magnetic field at center of a circular loop is proportional to
A) R B) $1/R$ C) R^2 D) $1/R^2$
8. For a perfect conductor, resistivity is
A) 1 B) ∞ C) 0 D) Depends on length
9. Photoelectric effect supports particle nature of
A) Electrons B) Protons C) Light D) Atoms
10. de Broglie wavelength is inversely proportional to
A) Velocity B) Momentum C) Energy D) Mass
11. Unit of self inductance is
A) Tesla B) Henry C) Weber D) Farad
12. In SHM, acceleration is proportional to
A) x B) $1/x$ C) x^2 D) constant
13. Young's modulus equals
A) Stress/Strain B) Strain/Stress C) Force/Area D) Area/Force
14. RMS value of AC current equals
A) I_0 B) $I_0/2$ C) $I_0/\sqrt{2}$ D) $\sqrt{2}I_0$
15. Induced emf is explained by
A) Ohm's law B) Faraday's law C) Coulomb's law D) Ampere's law
16. A body at rest has kinetic energy
A) 1 J B) ∞ C) 0 D) Depends on mass
17. Refractive index of vacuum is
A) 0 B) 1 C) ∞ D) Depends on wavelength
18. SI unit of capacitance is
A) Henry B) Farad C) Tesla D) Ohm
19. Electrostatic force varies as
A) r B) r^2 C) $1/r$ D) $1/r^2$

20. Energy of a photon is

A) hc/λ B) $h\lambda/c$ C) $c/h\lambda$ D) λ/hc

21. Area under velocity-time graph gives

A) Acceleration B) Distance C) Displacement D) Momentum

22. For an ideal gas

A) $PV=nRT$ B) $PV=RT/n$ C) $PV=n/R$ D) $PV=\text{constant}$

23. Nuclear force is

A) Electromagnetic B) Weak C) Strong D) Gravitational

24. A transformer works on

A) Photoelectric effect B) Mutual induction C) Electrolysis D) Conduction

25. Moment of inertia depends on

A) Mass distribution B) Only mass C) Only radius D) Velocity

Section B: 5 Numerical Questions

1. A force of 20 N acts on a body of mass 4 kg. Find acceleration (m/s^2).
2. A $10\ \Omega$ resistor carries 2 A current. Find potential difference (V).
3. Light of wavelength 500 nm travels in vacuum. Find frequency in units of 10^{14} Hz.
4. A body of mass 2 kg moves with speed 10 m/s. Find kinetic energy (J).
5. A gas expands from $2\ \text{m}^3$ to $5\ \text{m}^3$ at constant pressure 100 Pa. Find work done (J).